Question 1 [10 Marks]

A farmer wants to construct animal feeder from a metal sheet. He wants to optimize the design of the feeder to avoid wasting any material. The design of the animal feeder is a hollow open-topped half cylinder. The radius of the semicircular ends is r cm and the length of the feeder is L cm. The metal used in the construction of the feeder is $600\pi\ cm^2$.

a) Sketch the shape of the animal feeder.

b) What is the constraint equation?

c) Write length L in terms of radius r.

d) Write the equation of volume V in terms of r *and* L.

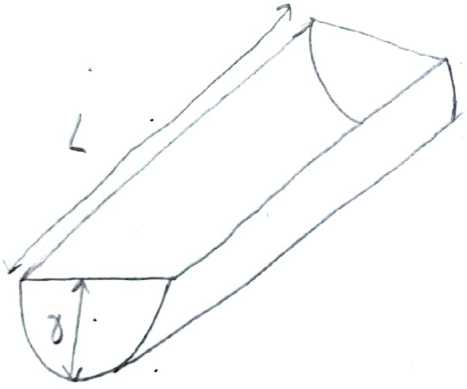
e) Write the equation of volume in terms of r alone.

f) Solve the optimization problem to determine the exact value of r for which V is optimized.

g) Verify that the value of r found in part (f) gives the maximum value for V.

h) Find, in exact form, the capacity and the length of the feeder.

(a)



(b) $A = 600\pi$

$\pi r^2 + \frac{1}{2}(2\pi rL) = 600\pi$

πr^2 ↓ 2 structures

$\frac{1}{2}(2\pi rL)$ ↓ curvey surface

$\pi r^2 + \pi rL = 600\pi$

(c) $r^2 + rL = 600$

$rL = 600 - r^2$

$$L = \frac{600 - r^2}{r}$$

(d) $V = \frac{1}{2}(\pi r^2 L) \rightarrow (i)$

(e) Substituting value of L in (i) we get

$$V = \frac{1}{2}\pi r^2 \left(\frac{600 - r^2}{r}\right)$$

$$V = \frac{\pi r^2 (600 - r^2)}{2r}$$

$$V = \frac{600\pi r}{2} - \frac{\pi r^3}{2}$$

$$V = 300\pi r - \frac{1}{2}\pi r^3 \rightarrow (ii)$$

(f) Diff eq (ii) w.r.t r

$$\frac{dV}{dr} = 300\pi - \frac{3}{2}\pi r^2$$

$$\frac{dV}{dr} = 0$$

$$300\pi - \frac{3}{2}\pi r^2 = 0$$

$$600\pi - 3r^2 = 0$$

$$600 = 3r^2$$

$$r^2 = 200$$

$$\boxed{r = 10\sqrt{2}} \qquad r > 0$$

(g) $$\frac{d^2V}{dr^2} = -3\pi r$$

$$\left.\frac{d^2V}{dr^2}\right|_{r = 10\sqrt{2}} = -30\pi\sqrt{2} < 0$$

It's a max according to 2nd derivative test.

(h) Substituting value of r in (ii) & (i) we get

$$V = 300\pi(10\sqrt{2}) - \frac{1}{2}\pi(10\sqrt{2})^3$$

$$V = 2000\pi\sqrt{2}\ cm^3$$

$$L = \frac{600 - (10\sqrt{2})^2}{10\sqrt{2}} \Rightarrow L = 20\sqrt{2}\ cm \quad , \quad r = 10\sqrt{2}\ cm$$

Question 2 [10 Marks]

In the engine shown below, a 7-inch connecting rod is fastened to a crank of radius 3 inches. The crankshaft rotates counterclockwise at a constant rate of 200 revolutions per minute.

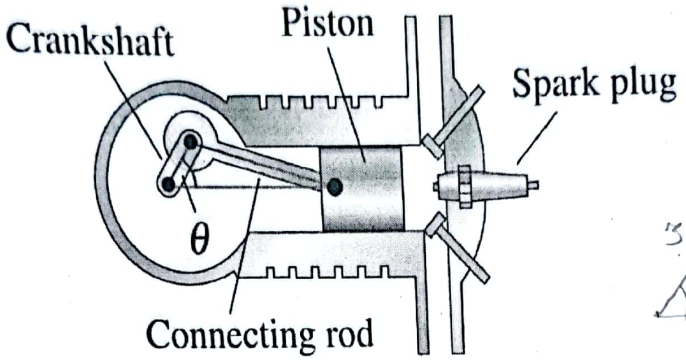
a) Write the expression for rate of change of θ.

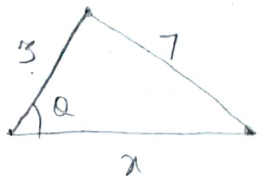
b) Find an equation that relates x and θ.

c) Find a differential equation for rate of change of position of the piston x.

d) Find position x for $\theta = \frac{\pi}{3}$.

e) Find the velocity of the piston at $\theta = \frac{\pi}{3}$.





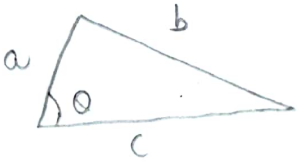
a) A complete revolution corresponds to 2π radians, it follows that

$$\frac{d\theta}{dt} = 200(2\pi)$$

$$\frac{d\theta}{dt} = 400\pi \quad \text{(constant rate)}$$

(b) Law of Cosine is used to find an eq that relates x & θ

$$7^2 = 3^2 + x^2 - 2(3)(x)\cos\theta \rightarrow (i)$$



$b^2 = a^2 + c^2 - 2ac\cos\theta$

c) Differentiating (i) w.r.t 't' we get

$$0 = 2x\frac{dx}{dt} - 6\left[-x\sin\theta\frac{d\theta}{dt} + \cos\theta\frac{dx}{dt}\right]$$

$$(6\cos\theta - 2x)\frac{dx}{dt} = 6x\sin\theta\frac{d\theta}{dt}$$

$$\frac{dx}{dt} = \frac{6x \sin\theta}{6\cos\theta - 2x}\left(\frac{d\theta}{dt}\right) \rightarrow (ii)$$

(d) Substituting value $\theta = \frac{\pi}{3}$ in (i) we get

$$7^2 = 3^2 + x^2 - 2(3)(x)\cos\frac{\pi}{3}$$

$$49 = 9 + x^2 - 6x\left(\frac{1}{2}\right)$$

$$0 = x^2 - 3x - 40$$

$$0 = (x-8)(x+5)$$

$$x = 8 \qquad \text{Choose +ive sol}$$

(e) Substitute $x = 8$ & $\theta = \frac{\pi}{3}$ in (ii) to find the velocity of piston

$$\frac{dx}{dt} = \frac{6(8)(\sqrt{3}/2)}{6\left(\frac{1}{2}\right) - 16}(400\pi)$$

$$= \frac{9600\pi\sqrt{3}}{-13}$$

$$\approx -4018 \text{ inches per minute}$$

* Velocity of the piston is -ive because x represents a distance that is decreasing.

Q4 Find $\frac{dy}{dx}$ of

$$x^y + y^x = c$$

Sol: Diff w.r.t 'x'

$$\frac{d}{dx}\left[x^y + y^x\right] = \frac{d}{dx}(c)$$

$$\frac{d}{dx}u + \frac{d}{dx}v = 0$$

$$x^y\left[\frac{y}{x} + \ln x \frac{dy}{dx}\right] + y^x\left[\frac{x}{y}\frac{dy}{dx} + \ln y\right] = 0$$

$$x^{y-1}\cdot y + x^y \ln x \frac{dy}{dx} + y^{x-1}x\frac{dy}{dx} + y^x \ln y = 0$$

$$\left(x^y \ln x + y^{x-1}x\right)\frac{dy}{dx} = -y^x \ln y - x^{y-1}y$$

$$\boxed{\frac{dy}{dx} = -\frac{y\left(y^{x-1}\ln y - x^{y-1}\right)}{x\left(x^{y-1}\ln x + y^{x-1}\right)}}$$ Ans

let $u = x^y$

$$\ln u = y \ln x$$

$$\frac{1}{u}\frac{du}{dx} = y\left(\frac{1}{x}\right) + \ln x \frac{dy}{dx}$$

$$\frac{du}{dx} = u\left[\frac{y}{x} + \ln x \frac{dy}{dx}\right]$$

Secondly

let $v = y^x$

$$\ln v = x \ln y$$

$$\frac{1}{v}\frac{dv}{dx} = x\left(\frac{1}{y}\right)\frac{dy}{dx} + \ln y$$

$$\frac{dv}{dx} = v\left[\frac{x}{y}\frac{dy}{dx} + \ln y\right]$$

Q#05 (a)

$$\frac{dE}{dm} = \frac{\text{Energy}}{\text{Time}} = 40$$

$$\frac{dm}{dt} = 8$$

$$\frac{dE}{dt} = \frac{dE}{dm} \times \frac{dm}{dt}$$

$$\Rightarrow \frac{dE}{dt} = 40 \times 8 = 320 \text{ calories.}$$

Q#05 (b)

By net change theorem

$$G = G_0 + \int_0^T g(t)\,dt$$

Where G_0 and G are Initial and Final Populations respectively.

By Integrating from $t=0$ to $T=36$ months we get the total increase in the number of people

$$\Rightarrow \int_0^T g(t)\,dt = \int_0^{36} (50 + 10\sqrt{t})\,dt = 50t + \frac{10t^{3/2}}{3/2}\Big|_0^{36}$$

$$= 50t + \frac{20}{3}t^{3/2}\Big|_0^{36}$$

$$= 3240$$

Thus, The final population in 3 years is expected to be $G = 5000 + 3240 = 8240$

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